Team 10
Marushika Manohar 2023103007
Suhasri S 2023103049
Harini J S 2023103549

TreeQA : Logic Tree Decomposition Framework for Multi-Hop Question Answering LLM-RAG System

Objective

- **Hierarchical Problem Decomposition**: Formulate complex multi-hop questions into structured logic trees using Pyramid and MECE principles to organize sub-questions and verifiable hypotheses.
- **Hybrid Context Retrieval**: Unify structured graph retrieval (Wikidata entity linking and relation filtering) and unstructured text retrieval (dense Wikipedia chunk search) for comprehensive evidence coverage.
- **Dynamic Self-Correction**: Implement a conflict-driven verification mechanism that identifies unverified or refuted hypotheses, invalidates faulty logic paths, and reconstructs valid sub-trees in real time.
- **Error Mitigation & Interpretability**: Eliminate compounding errors in multi-step reasoning while providing an interpretable, fully verified "golden tree" path leading to the final response.

Problem Statement

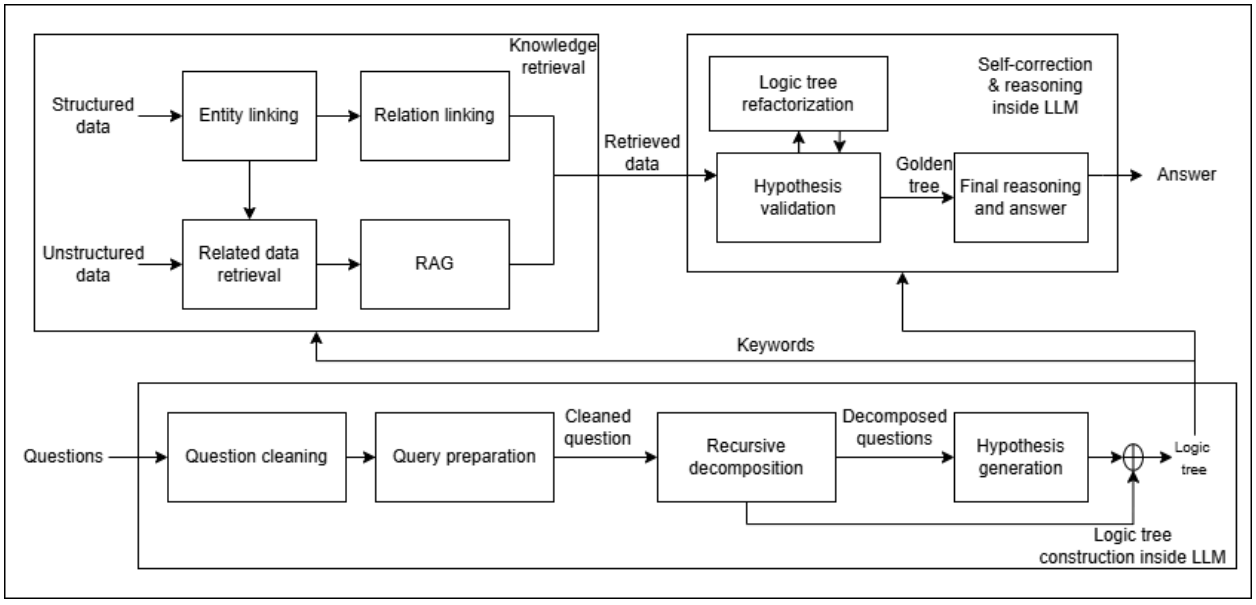
Current multi-hop QA approaches struggle to perform reliable complex reasoning because traditional sequential decomposition models accumulate errors across intermediate steps. When early sub-questions yield incorrect or missing answers, subsequent reasoning steps fail without any mechanism to detect or rectify the mistake. Furthermore, unconstructed retrieval over hybrid sources (structured knowledge graphs and unstructured text) often yields noisy context, leading to hallucinations or invalid logical leaps. The TreeQA framework addresses these limitations by explicitly modeling logical dependencies, verifying intermediate evidence, and dynamically recovering from reasoning errors.

Abstract

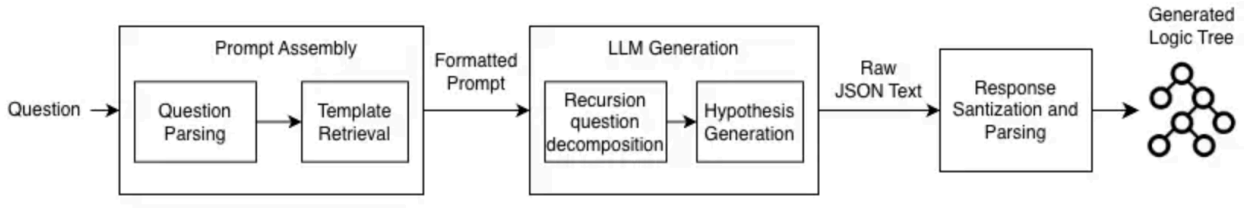
Multi-hop question answering (QA) requires reasoning across multiple scattered pieces of evidence to produce accurate answers. Existing methods often suffer from error propagation during sequential sub-question answering or lack structured logical reasoning. To address these challenges, the TreeQA framework structures complex reasoning into a hierarchical logic tree governed by the Pyramid Principle and MECE (Mutually Exclusive and Collectively Exhaustive)

guidelines. TreeQA dynamically decomposes root questions into actionable sub-hypotheses, retrieves grounding context from structured knowledge graphs and unstructured text, and performs dynamic self-correction by reconstructing invalid sub-trees upon encountering conflicting evidence. This project aims to implement and evaluate the TreeQA framework, demonstrating how its structured logic tree reduces compounding errors and enhances reasoning transparency across complex QA benchmarks.

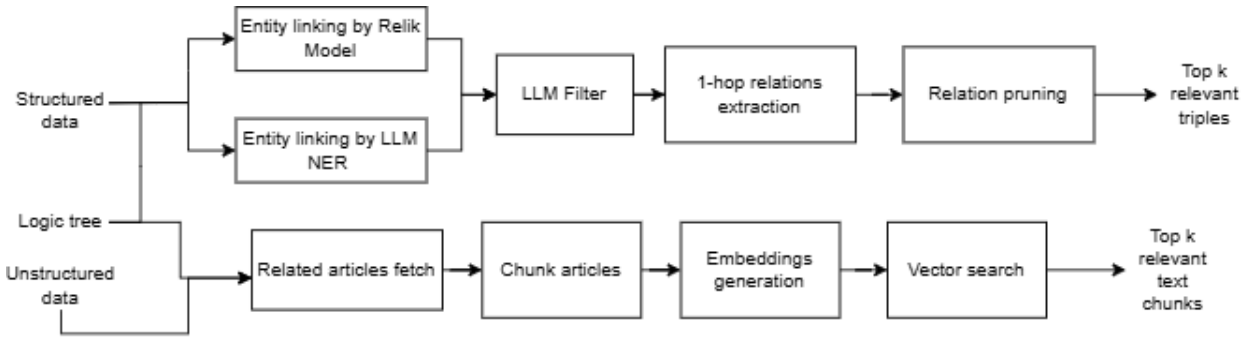
Overall Architecture Diagram



Logic Tree Construction Module Diagram



Knowledge Retrieval Module Diagram



Self Correction and Reasoning Module Diagram

